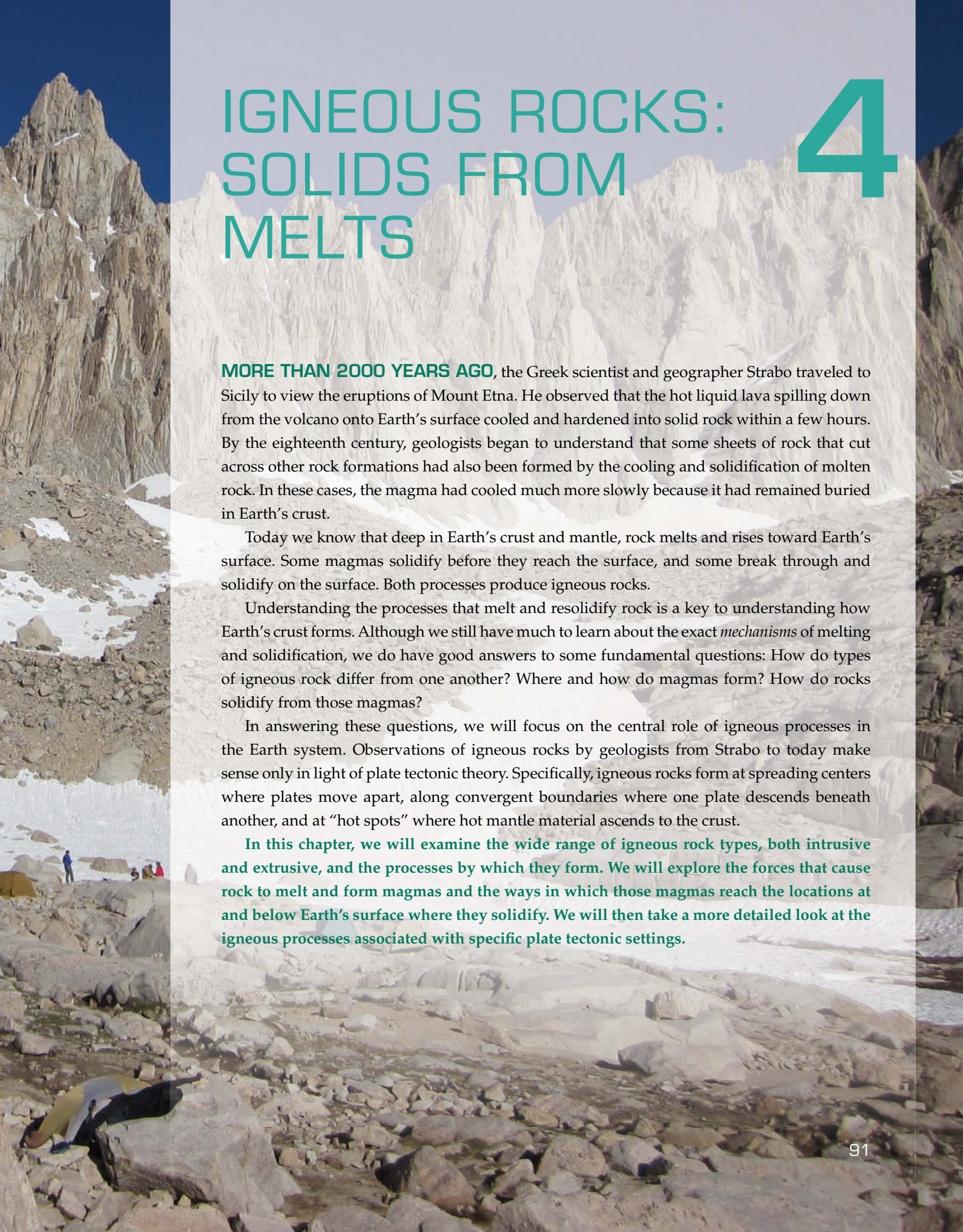




IGNEOUS ROCKS: SOLIDS FROM MELTS

4

MORE THAN 2000 YEARS AGO, the Greek scientist and geographer Strabo traveled to Sicily to view the eruptions of Mount Etna. He observed that the hot liquid lava spilling down from the volcano onto Earth's surface cooled and hardened into solid rock within a few hours. By the eighteenth century, geologists began to understand that some sheets of rock that cut across other rock formations had also been formed by the cooling and solidification of molten rock. In these cases, the magma had cooled much more slowly because it had remained buried in Earth's crust.

Today we know that deep in Earth's crust and mantle, rock melts and rises toward Earth's surface. Some magmas solidify before they reach the surface, and some break through and solidify on the surface. Both processes produce igneous rocks.

Understanding the processes that melt and resolidify rock is a key to understanding how Earth's crust forms. Although we still have much to learn about the exact *mechanisms* of melting and solidification, we do have good answers to some fundamental questions: How do types of igneous rock differ from one another? Where and how do magmas form? How do rocks solidify from those magmas?

In answering these questions, we will focus on the central role of igneous processes in the Earth system. Observations of igneous rocks by geologists from Strabo to today make sense only in light of plate tectonic theory. Specifically, igneous rocks form at spreading centers where plates move apart, along convergent boundaries where one plate descends beneath another, and at "hot spots" where hot mantle material ascends to the crust.

In this chapter, we will examine the wide range of igneous rock types, both intrusive and extrusive, and the processes by which they form. We will explore the forces that cause rock to melt and form magmas and the ways in which those magmas reach the locations at and below Earth's surface where they solidify. We will then take a more detailed look at the igneous processes associated with specific plate tectonic settings.

■ How Do Igneous Rocks Differ from One Another?

Geologists today classify igneous rock samples in the same way some geologists did in the late nineteenth century: by their texture and by their mineral and chemical composition.

Texture

Two hundred years ago, the first division of igneous rocks was made on the basis of texture, which largely reflects differences in mineral grain size: geologists classified rocks as either coarse-grained or fine-grained (see Chapter 3). Grain size is a simple characteristic that geologists can easily see in the field. A coarse-grained rock, such as granite, has distinct crystals that are easily visible to the naked eye. In contrast, the crystals of a fine-grained rock, such as basalt, are too small to be seen, even with a magnifying glass.

Figure 4.1 shows samples of granite and basalt, accompanied by photomicrographs of very thin, transparent slices of each rock. *Photomicrographs*, which are simply photographs taken through a microscope, give us an enlarged view of minerals and their textures. Textural differences were clear to early geologists, but several more clues were needed to unravel the meaning of those differences.

FIRST CLUE: VOLCANIC ROCKS Early geologists observed volcanic rocks forming from lava during volcanic eruptions. (**Lava** is the term that we apply to magma flowing out onto Earth's surface.) They noted that where lava cooled rapidly, it formed either a fine-grained rock or a glassy one in which no crystals could be distinguished. Where lava

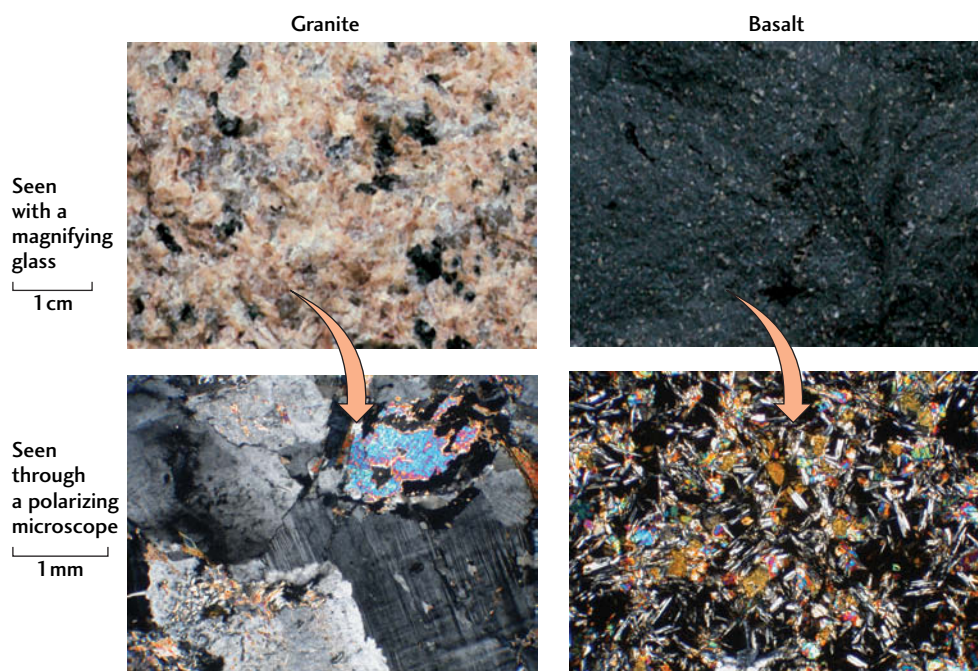
cooled more slowly, as in the middle of a thick flow many meters high, somewhat larger crystals were formed.

SECOND CLUE: LABORATORY STUDIES OF CRYSTALLIZATION Just over a hundred years ago, experimental scientists began to understand the nature of crystallization. Anyone who has frozen a tray of ice cubes knows that water solidifies to ice in a few hours as its temperature drops below the freezing point. If you have ever attempted to retrieve your ice cubes before they were completely solid, you may have seen thin ice crystals forming at the surface and along the sides of the tray. During crystallization, the water molecules take up fixed positions in the solidifying crystal structure, and they are no longer able to move freely, as they did when the water was liquid. All other liquids, including magmas, crystallize in this way.

The first tiny crystals form a pattern. Other atoms or ions in the crystallizing liquid then attach themselves in such a way that the tiny crystals grow larger. It takes some time for the atoms or ions to “find” their correct places on a growing crystal, so crystals grow large only if they have time to grow slowly. If a liquid solidifies very quickly, as a magma does when it erupts onto the cool surface of Earth, the crystals have no time to grow. Instead, a large number of tiny crystals form simultaneously as the liquid cools and solidifies.

THIRD CLUE: GRANITE AS EVIDENCE OF SLOW COOLING By studying volcanoes, early geologists determined that fine-grained textures indicate quick cooling at Earth's surface and that fine-grained igneous rocks are evidence of former volcanism. But in the absence of direct observation, how could geologists deduce that coarse-grained rocks form by *slow* cooling deep in Earth's interior? Granite—

FIGURE 4.1 ■ Igneous rocks were first classified by texture. Early geologists assessed rock texture with a small hand-held magnifying glass. Modern geologists have access to high-powered polarizing microscopes, which can produce photomicrographs of thin, transparent rock slices like those shown here. [Photos by John Grotzinger/Ramón Rivera-Moret/Harvard Mineralogical Museum; photomicrographs by Steven Chemtob.]



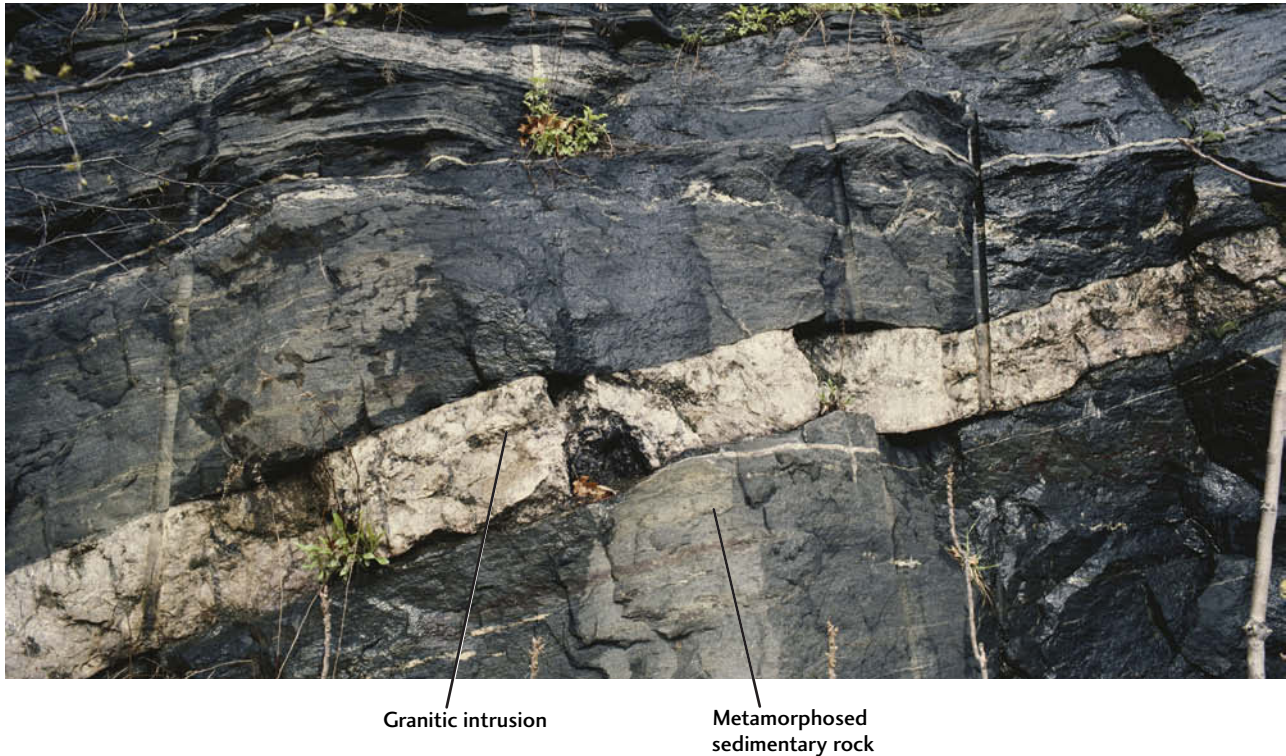


FIGURE 4.2 ■ Granite pegmatite sill or dike (the lighter-colored rock) in an outcrop of schist (darker colored rock) along the Harlem River, New York, suggests to geologists that the intruding rock had been forced into the fractures as a liquid. [Catherine Ursillo/Science Source.]

one of the commonest rocks of the continents—turned out to be the crucial clue (**Figure 4.2**). James Hutton, one of geology's founding fathers, saw granite cutting across and disrupting layers of sedimentary rock as he worked in the field in Scotland. He noticed that the granite had somehow fractured and invaded the sedimentary rock, as though the granite had been forced into the fractures as a liquid.

As Hutton looked at more and more granites, he began to focus on the sedimentary rocks bordering them. He observed that the minerals of the sedimentary rocks in contact with the granite were different from those found in sedimentary rocks at some distance from the granite. He concluded that the changes in the sedimentary rocks must have resulted from great heat, and that the heat must have come from the granite. Hutton also noted that the granite was composed of interlocked crystals (see **Figure 4.1**). By this time, chemists had established that a slow crystallization process produces this pattern.

With these three lines of evidence, Hutton proposed that granite forms from hot molten material that solidifies deep within Earth. The evidence was conclusive because no other explanation could accommodate all the facts. Other geologists, who saw the same characteristics of granites in widely separated places throughout the world, came to recognize that granite and many similar coarse-grained rocks were the products of magma that had crystallized slowly in Earth's interior.

INTRUSIVE AND EXTRUSIVE TEXTURES The full significance of an igneous rock's texture is now clear: it is linked to the rate, and therefore the place, of cooling. An **intrusive igneous rock** is one that has forced its way into the surrounding rock, called **country rock**, and solidified without reaching Earth's surface. Slow cooling of magma in Earth's interior allows adequate time for the growth of the large, interlocking crystals that characterize intrusive igneous rocks (**Figure 4.3**).

Rapid cooling at Earth's surface produces the fine-grained texture or glassy appearance of **extrusive igneous rocks** (see **Figure 4.3**). These rocks, formed partly or largely of volcanic glass, are formed from material that erupts from volcanoes. For this reason, they are also known as *volcanic rocks*. They fall into two major categories based on the type of erupted material from which they are formed:

- **Lavas:** Volcanic rocks formed from flowing lavas range in appearance from smooth and ropy to sharp, spiky, and jagged, depending on the conditions under which they are formed.
- **Pyroclasts:** In more violent eruptions, **pyroclasts** form when fragments of lava are thrown high into the air. **Volcanic ash** is made up of extremely small fragments, usually of glass, that form when escaping gases force a fine spray of magma from a volcano. **Bombs** are larger particles hurled from the volcano

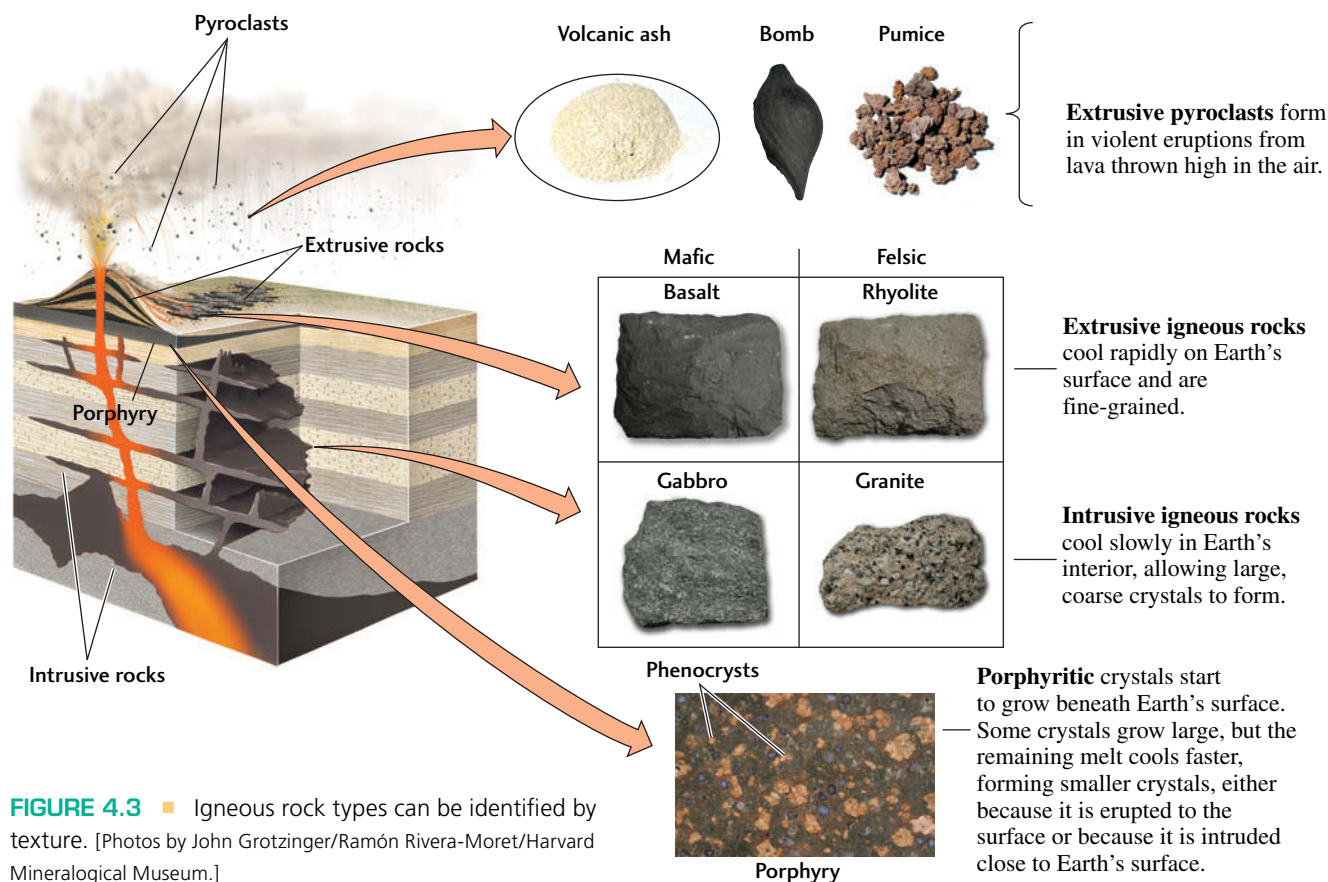


FIGURE 4.3 ■ Igneous rock types can be identified by texture. [Photos by John Grotzinger/Ramón Rivera-Moret/Harvard Mineralogical Museum.]

and streamlined by the air as they hurtle through it. As they fall to the ground and cool, these fragments of volcanic debris may stick together to form rocks.

One volcanic rock type is **pumice**, a frothy mass of volcanic glass in which a great number of spaces remain after trapped gas has escaped from the solidifying melt. Another wholly glassy volcanic rock type is **obsidian**; unlike pumice, it contains only tiny vesicles and so is solid and dense. Chipped or fragmented obsidian produces very sharp edges, and Native Americans and many other hunting groups used it for arrowheads and a variety of cutting tools.

A **porphyry** is an igneous rock that has a mixed texture in which large crystals “float” in a predominantly fine-grained matrix (see Figure 4.3). The large crystals, called *phenocrysts*, form in magma while it is still below Earth's surface. Then, before other crystals can grow, a volcanic eruption brings the magma to the surface, where it cools quickly to a finely crystalline mass. In some cases, porphyries form as intrusive igneous rocks; for example, they may form where magmas cool quickly at very shallow levels in the crust. Porphyry textures are important to geologists because they show that different minerals crystallize at different rates, a point that will be emphasized later in this chapter.

In Chapter 12, we will look more closely at how volcanic processes form extrusive igneous rocks. Now, however,

we turn to the second way in which the family of igneous rocks is subdivided.

Chemical and Mineral Composition

We have just seen how igneous rocks can be subdivided according to their texture. They can also be classified on the basis of their chemical and mineral composition. Volcanic glass, which is formless even under a microscope, is often classified by chemical analysis alone. One of the earliest classifications of igneous rocks was based on a simple chemical analysis of their silica content. Silica (SiO_2) is abundant in most igneous rocks, accounting for 40 to 70 percent of their total weight.

Modern classifications group igneous rocks according to their relative proportions of silicate minerals (Table 4.1; see also Appendix 4).

The silicate minerals—quartz, feldspars, muscovite and biotite micas, amphiboles and pyroxenes, and olivine—form a systematic series. *Felsic* minerals are the highest in silica; *mafic* minerals are the lowest in silica. The adjectives *felsic* (from *feldspar* and *silica*) and *mafic* (from *magnesium* and *ferric*, from the Latin *ferrum*, “iron”) are applied both to minerals and to rocks containing large proportions of those minerals. Mafic minerals crystallize at higher

TABLE 4-1 Common Minerals of Igneous Rocks

Compositional Group	Mineral	Chemical Composition	Silicate Structure
FELSIC	Quartz	SiO_2	Frameworks
	Orthoclase feldspar	KAlSi_3O_8	
	Plagioclase feldspar	$\text{NaAlSi}_3\text{O}_8$; $\text{CaAl}_2\text{Si}_2\text{O}_8$	
	Muscovite (mica)	$\text{KAl}_3\text{Si}_3\text{O}_{10}(\text{OH})_2$	
MAFIC	Biotite (mica)	K Mg Fe Al } $\text{Si}_3\text{O}_{10}(\text{OH})_2$	Double chains
	Amphibole group	Mg Fe Ca Na } $\text{Si}_8\text{O}_{22}(\text{OH})_2$	
	Pyroxene group	Mg Fe Ca Al } SiO_3	
	Olivine	$(\text{Mg,Fe})_2\text{SiO}_4$	
			Single chains
			Isolated tetrahedral

temperatures—that is, earlier in the cooling of a magma—than felsic minerals.

As the mineral and chemical compositions of igneous rocks became known, geologists soon noticed that some extrusive and intrusive rocks were identical in composition and differed only in texture. Basalt, for example, is an extrusive rock formed from lava. Gabbro has exactly the same mineral and chemical composition as basalt, but forms deep in Earth's crust (see Figure 4.3). Similarly, rhyolite and granite are identical in composition, but differ in texture. Thus, extrusive and intrusive rocks form two chemically and mineralogically parallel sets of igneous rocks. Conversely, most of the chemical and mineral compositions in the felsic-to-mafic series we have just described can appear in either extrusive or intrusive rocks. The only exceptions are very highly mafic rocks, which rarely appear as extrusive igneous rocks.

Figure 4.4 is a model that portrays these relationships. The horizontal axis plots silica content as a percentage of a given rock's weight. The percentages given—from high silica content at 70 percent to low silica content at 40 percent—cover the range found in igneous rocks. The vertical axis plots mineral content as a percentage of a given

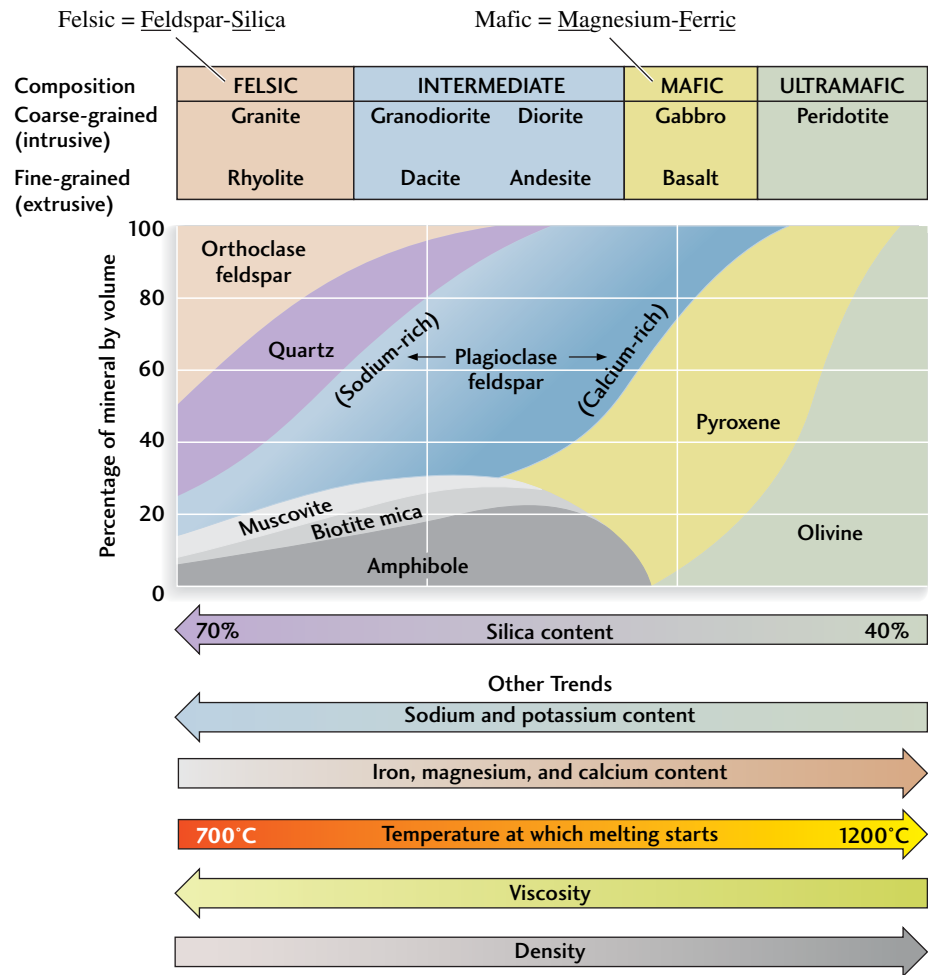
rock's volume. This model can be used to classify an unknown rock sample with a known silica content: by finding its silica content on the horizontal axis, you can determine its mineral composition and, from that, the type of rock it is.

We can use Figure 4.4 to guide our discussion of intrusive and extrusive igneous rocks. We begin with the felsic rocks at the far left of the model.

FELSIC ROCKS Felsic rocks are poor in iron and magnesium and rich in felsic minerals that are high in silica. Such minerals include quartz, orthoclase feldspar, and plagioclase feldspar. Orthoclase feldspars, which contain potassium, are more abundant than plagioclase feldspars. Plagioclase feldspars contain varying amounts of calcium and sodium; as Figure 4.4 indicates, they are richer in sodium near the felsic end and richer in calcium near the mafic end of the scale. Thus, just as mafic minerals crystallize at higher temperatures than felsic minerals, calcium-rich plagioclases crystallize at higher temperatures than sodium-rich plagioclases.

Felsic rocks tend to be light in color. **Granite**, one of the most abundant intrusive igneous rocks, contains about 70 percent silica. Its mineral composition includes abundant

FIGURE 4.4 ■ Classification model for igneous rocks. The vertical axis shows the minerals contained in a given rock as a percentage of its volume. The horizontal axis shows the silica content of a given rock as a percentage of its weight. Thus, if you knew by chemical analysis that a coarsely textured rock sample was about 70 percent silica, you could deduce that its composition was about 6 percent amphibole, 3 percent biotite, 5 percent muscovite, 14 percent plagioclase feldspar, 22 percent quartz, and 50 percent orthoclase feldspar. Your rock would be granite. Although rhyolite has the same mineral composition, its fine texture would eliminate it from consideration.



quartz and orthoclase feldspar and a smaller amount of plagioclase feldspar (see the far left of Figure 4.4). These light-colored felsic minerals give granite its pink or gray color. Granite also contains small amounts of muscovite and biotite micas and amphibole. **Rhyolite** is the extrusive equivalent of granite. This light brown to gray rock has the same felsic composition and light coloration as granite, but it is much more fine-grained. Many rhyolites are formed largely or entirely of volcanic glass.

INTERMEDIATE IGNEOUS ROCKS Midway between the felsic and mafic ends of the scale are the **intermediate igneous rocks**. As their name indicates, these rocks are neither as rich in silica as the felsic rocks nor as poor in it as the mafic rocks. We find the intermediate intrusive igneous rocks to the right of granite in Figure 4.4. The first is **granodiorite**, a light-colored rock that looks something like granite. It is also similar to granite in having abundant quartz, but its predominant feldspar is plagioclase, not orthoclase. To its right is **diorite**, which contains still less silica and is dominated by plagioclase feldspar, with little or no quartz. Diorites contain a moderate amount of the mafic

minerals biotite, amphibole, and pyroxene. They tend to be darker than granite or granodiorite.

The volcanic equivalent of granodiorite is **dacite**. To its right in the extrusive series is **andesite**, the volcanic equivalent of diorite. Andesite derives its name from the Andes, the volcanic mountain belt in South America.

MAFIC ROCKS **Mafic rocks** contain large proportions of pyroxenes and olivines. These minerals are relatively poor in silica but are rich in magnesium and iron, from which they get their characteristic dark colors. **Gabbro** is a coarse-grained, dark gray intrusive igneous rock. Gabbro has an abundance of mafic minerals, especially pyroxenes. It contains no quartz and only moderate amounts of calcium-rich plagioclase feldspar.

Basalt is the most abundant igneous rock of the crust, and it underlies virtually the entire seafloor. This dark gray to black rock is the fine-grained extrusive equivalent of gabbro. In some places, extensive thick sheets of basalt, called *flood basalts*, form large plateaus. The Columbia River basalts of Washington State and the remarkable formation known as the Giant's Causeway in Northern Ireland are

two examples. The Deccan flood basalts of India and the Siberian flood basalts of northern Russia were formed by enormous outpourings of basalt that appear to coincide closely with two of the greatest periods of mass extinction in the fossil record.

ULTRAMAFIC ROCKS Ultramafic rocks consist primarily of mafic minerals and contain less than 10 percent feldspar. Here, at the far right of Figure 4.4, with a silica content of only about 45 percent, we find **peridotite**, a coarse-grained, dark greenish gray rock made up primarily of olivine with smaller amounts of pyroxene. Peridotites are the dominant rocks in Earth's mantle, and as we will see, they are the source of the basaltic magmas that form rocks at mid-ocean ridges. Ultramafic rocks are rarely found as extrusives. Because they solidify at such high temperatures, they are rarely liquid and hence do not form typical lavas.

TRENDS IN THE FELSIC-TO-MAFIC SERIES The names and exact compositions of the various rocks in the felsic-to-mafic series are less important to remember than the trends shown in Figure 4.4. There is a strong correlation between a rock's mineralogy and its temperature of crystallization or melting. As Table 4.2 indicates, mafic minerals melt at higher temperatures than felsic minerals. At temperatures below their melting point, minerals crystallize; therefore, mafic minerals also crystallize at higher temperatures than felsic minerals. We can also see that silica content increases as we move from the mafic end to the felsic end of the series. Increasing silica content results in increasingly complex silicate structures (see Table 4.1), which interfere with a melted rock's ability to flow. Thus, **viscosity**—the measure of a liquid's *resistance* to flow—increases as silica content increases. Viscosity is an important factor in the behavior of lavas, as we will see in Chapter 12. Increasing silica content also results in decreasing density, as we saw in Chapter 1.

It is clear that an igneous rock's mineralogy provides a great deal of information about the conditions under which the rock's parent magma formed and crystallized. To interpret this information accurately, however, we must understand more about igneous processes. We turn to that topic next.

TABLE 4-2 Factors Affecting Melting Temperatures

Higher Melting Temperatures	Lower Melting Temperatures
← Pressure increasing Water content increasing → Rock Composition ← More mafic More felsic →	

■ How Do Magmas Form?

We know from the way Earth transmits seismic waves that the bulk of the planet is solid for thousands of kilometers down to the core-mantle boundary (see Chapter 1). The evidence of volcanic eruptions, however, tells us that there must also be liquid regions where magmas originate. How do we resolve this apparent contradiction? The answer lies in the processes that melt rocks and create magmas.

How Do Rocks Melt?

Although we do not yet understand the exact mechanisms of rock melting and solidification within Earth, we have learned a great deal from laboratory experiments using high-temperature furnaces (Figure 4.5). From these

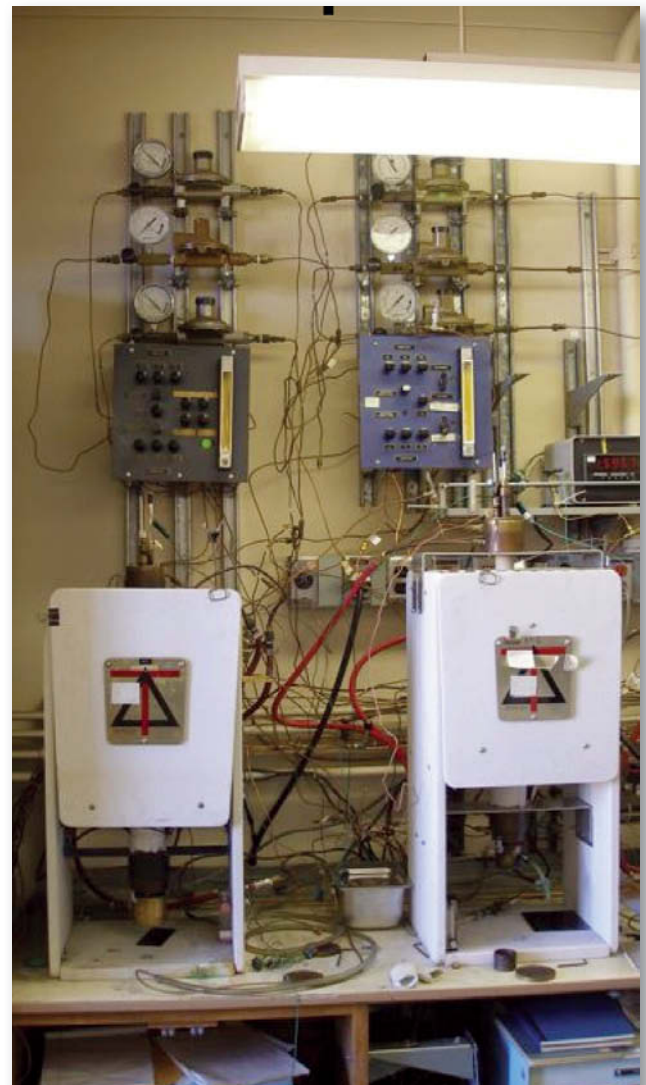


FIGURE 4.5 ■ Experimental device used to melt rocks in laboratory. [Sally Newman.]

experiments, we know that a rock's melting point depends on its chemical and mineral composition and on conditions of temperature and pressure (see Table 4.2).

TEMPERATURE AND MELTING A hundred years ago, geologists discovered that rock does not melt completely at a given temperature. Instead, rocks undergo **partial melting** because the minerals that compose them melt at different temperatures. As temperatures rise, some minerals melt and others remain solid. If the same conditions are maintained at any given temperature, the same mixture of solid and melted rock is maintained. The fraction of rock that has melted at a given temperature is called a *partial melt*. To visualize a partial melt, think of how a chocolate chip cookie would look if you heated it to the point at which the chocolate chips melted while the main part of the cookie stayed solid. The chips represent the partial melt, or magma.

The ratio of solid to partial melt depends on the proportions and melting temperatures of the minerals that make up the original rock. It also depends on the temperature at the depth in the crust or mantle where melting takes place. At the lower end of a rock's melting range, a partial melt might be less than 1 percent of the volume of the original rock. Much of the hot rock would still be solid, but significant amounts of liquid would be present as small droplets in the tiny spaces between crystals throughout the mass. In the upper mantle, for example, some basaltic magmas are produced by only 1 to 2 percent melting of peridotite. However, 15 to 20 percent melting of mantle peridotite to form basaltic magmas is common beneath mid-ocean ridges. At the high end of a rock's melting range, much of the rock would be liquid, containing lesser amounts of unmelted crystals. An example would be the reservoir of basaltic magma and crystals just beneath a volcano such as the island of Hawaii. Geologists have used this knowledge of partial melts to determine how different kinds of magmas form at different temperatures and in different regions of Earth's interior. As you can imagine, the composition of a magma formed from completely melted rock may be very different from that of a magma formed from rock in which only the minerals with the lowest melting points have melted. Thus, basaltic magmas that form in different regions of the mantle may have somewhat different compositions.

PRESSURE AND MELTING To get the whole story on melting, we must consider pressure as well as temperature. Pressure increases with depth within Earth as a result of the increasing weight of overlying rock. Geologists found that as they melted rocks under various pressures in the laboratory, higher pressures led to higher melting temperatures. Thus, rocks that would melt at a given temperature at Earth's surface would remain solid at the same temperature in Earth's interior. For example, a rock that melts at 1000°C at Earth's surface might have a much higher melting temperature, perhaps 1300°C, deep in the interior, where

pressures are many thousands of times greater than those at the surface. It is the effect of pressure that explains why the rocks in most of the crust and mantle do not melt. Rock can melt only when both temperature and pressure conditions are right.

Just as an increase in pressure can keep rock solid, a decrease in pressure can make rock melt, given a sufficiently high temperature. Because of convection currents in the mantle, mantle material rises to Earth's surface at mid-ocean ridges at a more or less constant temperature. As the material rises and the pressure on it decreases below a critical point, the solid rock melts spontaneously, without the introduction of any additional heat. This process, known as **decompression melting**, produces the greatest volume of magma anywhere on Earth. It is the process by which most basalts form on the seafloor.

WATER AND MELTING The many experiments on melting temperatures and partial melting of rocks paid other dividends as well. One of them was a better understanding of the role of water in melting. Geologists studying natural lavas in the field determined that water was present in some magmas. This finding gave them the idea of adding water to their experimental melts back in the laboratory. By adding small but varying amounts of water, they discovered that the compositions of partial melts varied not only with temperature and pressure, but also with the amount of water present.

Consider, for example, the effect of dissolved water on pure albite, a sodium-rich plagioclase feldspar, at the low pressures at Earth's surface. If only a small amount of water is present in the rock, the rock will remain solid at temperatures just over 1000°C, hundreds of degrees above the boiling point of water. At these temperatures, the water in the albite is present as a vapor (gas). If large amounts of water are dissolved in the albite, however, its melting temperature will decrease, dropping to as low as 800°C. This behavior follows the general rule that dissolving one substance (in this case, water vapor) in another (in this case, albite) lowers the melting temperature of the solution. If you live in a cold climate, you are probably familiar with this principle because you know that salt sprinkled on icy roads lowers the melting temperature of the ice. By the same principle, the melting temperature of albite—and of all silicate minerals—drops considerably in the presence of large amounts of water. The melting points of these minerals decrease in proportion to the amount of water dissolved in the molten silicate.

Melting of rock induced by the presence of water that lowers its melting point is referred to as **fluid-induced melting**. Water content is a significant factor in the melting of sedimentary rocks, which contain an especially large volume of water in their pore spaces, more than is found in igneous or metamorphic rocks. As we will see later in this chapter, the water in sedimentary rocks plays an important role in the melting that gives rise to much of the volcanic activity at subduction zones.

The Formation of Magma Chambers

Most substances are less dense in their liquid form than in their solid form. The density of melted rock is lower than the density of solid rock of the same composition. With this knowledge, geologists reasoned that large bodies of magma could form in the following way: If the less dense melted rock were given a chance to move, it would move upward—just as oil, which is less dense than water, rises to the surface of a mixture of oil and water. Being liquid, a partial melt could move slowly upward through pores and along the boundaries between crystals of the surrounding solid rock. As the hot drops of melted rock moved upward, they would mix with other drops, gradually forming larger pools of magma within Earth's solid interior.

The rise of magmas through the mantle and crust may be slow or rapid. Magmas rise at rates from 0.3 m/year to almost 50 m/year, over periods of tens of thousands or even hundreds of thousands of years. As they ascend, magmas may mix with other melts and may also melt portions of the crust. We now know that large pools of molten rock, called **magma chambers**, form in the lithosphere as rising magmas melt and push aside surrounding solid rock. We know that they exist because seismic waves have shown us the depth, size, and general outlines of the magma chambers underlying some active volcanoes. A magma chamber may encompass a volume as large as several cubic kilometers. We cannot yet say exactly how magma chambers form, nor exactly what they look like in three dimensions. We can think of them as large, liquid-filled cavities in solid rock, which expand as more of the surrounding rock melts or as magma migrates through cracks and other small openings. Magma chambers contract as they expel magma to the surface in volcanic eruptions.

Where Do Magmas Form?

Our understanding of igneous processes stems from geologic inferences as well as laboratory experimentation. One important source of information is volcanoes, which give us information about where magmas are located. Another is the record of temperatures measured in deep drill holes and mine shafts. This record shows that the temperature of Earth's interior increases with depth. Using these measurements, scientists have been able to estimate the rate at which temperature rises as depth increases.

The temperatures recorded at a given depth in some locations are much higher than the temperatures recorded at the same depth in other locations. These results indicate that some parts of Earth's mantle and crust are hotter than others. For example, the Great Basin of the western United States is an area where the North American continent is being stretched and thinned, with the result that the temperature increases with depth at an exceptionally rapid rate, reaching 1000°C at 40 km, not far below the base of the crust. This temperature is almost high enough to melt

basalt. By contrast, in tectonically stable regions, such as the central parts of continents, the temperature increases much more slowly, reaching only 500°C at the same depth.

■ Magmatic Differentiation

The processes we've discussed so far account for the melting of rocks to form magmas. But what accounts for the variety of igneous rocks? Are magmas of different chemical compositions made by the melting of different kinds of rock? Or do igneous processes produce a variety of rocks from an originally uniform parent material?

Again, the answers to these questions came from laboratory experiments. Geologists mixed chemical elements in proportions that simulated the compositions of natural igneous rocks, then melted those mixtures. As the melts cooled and solidified, the geologists observed and recorded the temperatures at which crystals formed, as well as the chemical compositions of those crystals. This research gave rise to the theory of **magmatic differentiation**, a process by which rocks of varying composition can arise from a uniform parent magma. Magmatic differentiation occurs because different minerals crystallize at different temperatures.

In a kind of mirror image of partial melting, the last minerals to melt are the first minerals to crystallize from a cooling magma. This initial crystallization withdraws chemical elements from the melt, changing the magma's composition. Continued cooling crystallizes the minerals that melted at the next lower temperature range. Again, the magma's chemical composition changes as various elements are withdrawn. Finally, as the magma solidifies completely, the last minerals to crystallize are the ones that melted first. Thus, the same parent magma, because of its changing chemical composition throughout the crystallization process, can give rise to different types of igneous rocks.

Fractional Crystallization: Laboratory and Field Observations

Fractional crystallization is the process by which the crystals formed in a cooling magma are segregated from the remaining liquid rock. This segregation happens in several ways, following a sequence commonly described as *Bowen's reaction series* (Figure 4.6). In the simplest scenario, crystals formed in a magma chamber settle to the chamber's floor and are thus removed from further reaction with the remaining liquid. Thus, crystals that form early are segregated from the remaining magma, which continues to crystallize as it cools.

The effects of fractional crystallization can be seen in the Palisades, a line of imposing cliffs that faces the city of New York on the west bank of the Hudson River (Figure 4.7). This igneous formation is about 80 km long and, in places, more than 300 m high. It formed as a magma of basaltic composition intruded into nearly horizontal layers of

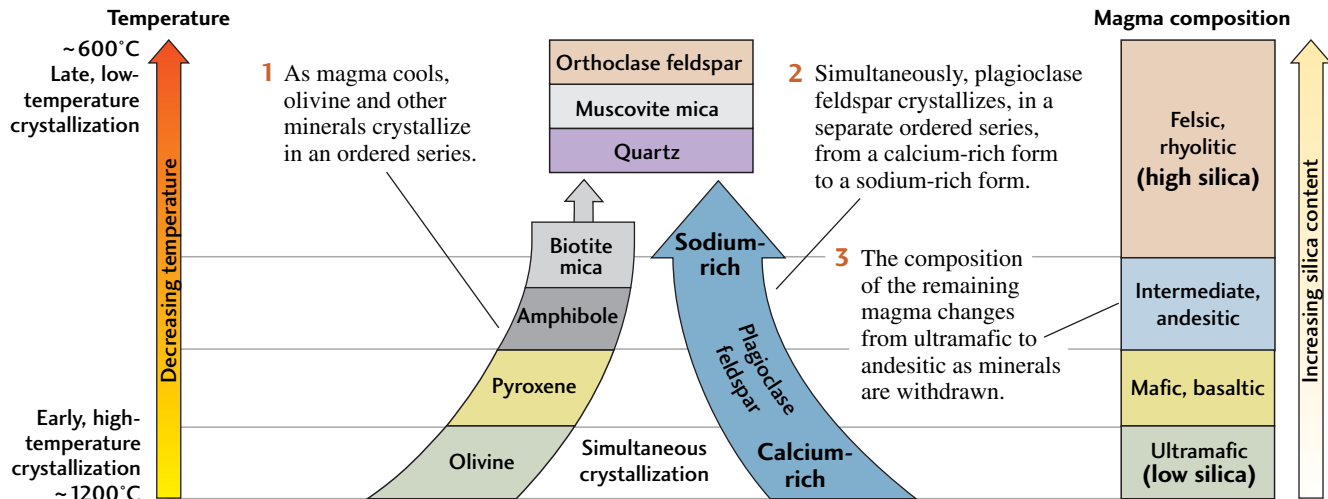


FIGURE 4.6 ■ Bowen's reaction series provides a model of fractional crystallization.

sedimentary rock. It contains abundant olivine near the bottom, pyroxene and calcium-rich plagioclase feldspar in the middle, and mostly sodium-rich plagioclase feldspar near the top. This variation in mineral composition from bottom to top made the Palisades a perfect site for testing the theory of fractional crystallization.

Geologists melted rocks with about the same mineral compositions as those found in the Palisades intrusion and determined that the initial temperature of the magma from which it formed had to have been about 1200°C. The parts of the magma within a few meters of the relatively cold country rock above and below it cooled quickly. This quick cooling formed a fine-grained basalt and preserved the

chemical composition of the original magma. The hot interior cooled more slowly, as evidenced by the slightly larger crystals found in the intrusion's interior.

The theory of fractional crystallization leads us to expect that the first mineral to crystallize from the slowly cooling interior of the Palisades intrusion would have been olivine, as this heavy mineral would sink through the melt to the bottom of the intrusion. It can be found today as a coarse-grained, olivine-rich layer just above the chilled, fine-grained basaltic layer along the bottom zone of contact with the underlying sedimentary rock. Plagioclase feldspar would have started to crystallize at about the same time; it has a lower density than olivine, however, and so

As magma cools, minerals crystallize at different temperatures and settle out of the magma in a particular order.

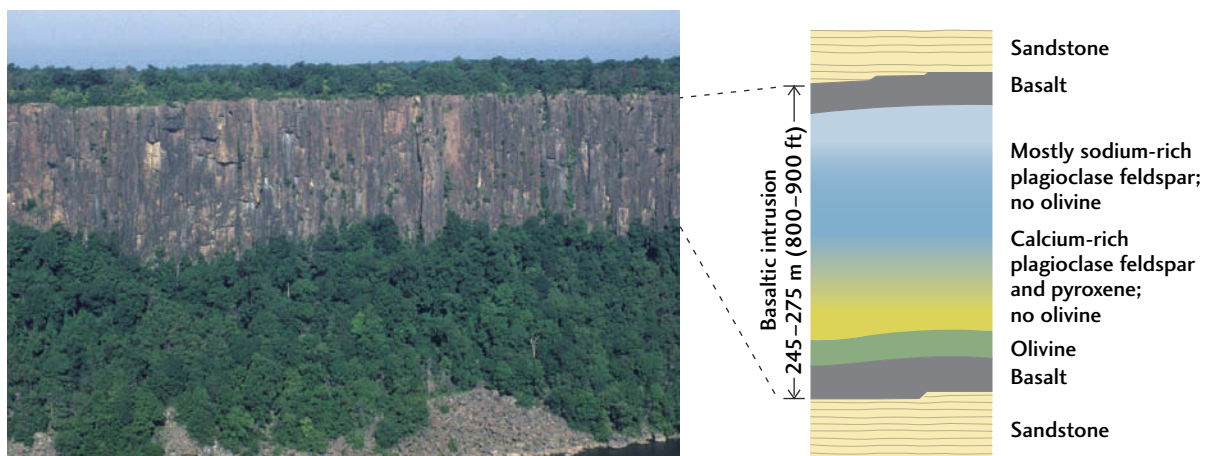


FIGURE 4.7 ■ Fractional crystallization explains the composition of the basaltic intrusion that forms the Palisades. Minerals in the Palisades intrusion are ordered with olivine at the bottom, a gradient of pyroxene and calcium-rich plagioclase feldspar in the center, and sodium-rich plagioclase feldspar at the top. Layers of fine-grained basalt, which cooled quickly at the edges of the intrusion, surround the more slowly cooled interior. [© Breck Kent.]

would have settled to the bottom more slowly (see Practicing Geology). Continued cooling would have produced pyroxene crystals, which would have reached the bottom next, followed almost immediately by calcium-rich plagioclase feldspar. The abundance of plagioclase feldspar in the upper parts of the intrusion is evidence that the magma continued to change in composition until successive layers of settled crystals were topped off by a layer of mostly sodium-rich plagioclase feldspar. In addition to crystallizing at a lower temperature, sodium-rich plagioclase feldspar is less dense than either olivine or pyroxene, so it would have settled out last.

Being able to explain the layering of the Palisades intrusion as the result of fractional crystallization was an early success in understanding magmatic differentiation. It firmly tied field observations to laboratory results and was solidly based on chemical knowledge. We now know that this intrusion actually has a more complex history that includes several injections of magma and a more complicated process of olivine settling. Nevertheless, the Palisades intrusion remains a valid example of fractional crystallization.

Granite from Basalt: Complexities of Magmatic Differentiation

Studies of volcanic lavas have shown that basaltic magmas are common—far more common than the rhyolitic magmas that correspond in composition to granites. How, then, could granites have become so abundant in Earth's crust? The answer is that the process of magmatic differentiation is much more complex than geologists first thought.

The original theory of magmatic differentiation suggested that a basaltic magma would gradually cool and differentiate into a more felsic magma by fractional crystallization. The early stages of this differentiation would produce an andesitic magma, which might erupt to form andesitic lavas or solidify by slow crystallization to form dioritic intrusions. Intermediate stages would result in magmas of granodioritic composition. If the process were carried far enough, its late stages would form rhyolitic lavas and granitic intrusions. One line of research has shown, however, that so much time would be needed for small crystals of olivine to settle through a dense, viscous magma that they might never reach the bottom of a magma chamber. Other researchers have demonstrated that many layered intrusions—similar to but much larger than the Palisades intrusion—do not show the simple progression of layers predicted by the original theory.

The greatest sticking point in the original theory, however, was the source of granite. The great volume of granite found on Earth could not have formed from basaltic magmas by magmatic differentiation, because large quantities of liquid volume are lost by crystallization during successive stages of differentiation. To produce a given amount of granite, an initial volume of basaltic magma 10 times that of the granitic intrusion would be required. Based on that

observation, there should be huge quantities of basalt underlying granitic intrusions. But geologists could not find anything like that amount of basalt. Even where great volumes of basalt are found—at mid-ocean ridges—there is no wholesale conversion into granite through magmatic differentiation.

Most in question is the original idea that all granitic rocks are formed from the differentiation of a single type of magma, a basaltic melt. Instead, geologists now believe that the melting of varied rock types in the upper mantle and crust is responsible for much of the observed variation in the composition of magmas:

1. Rocks in the upper mantle undergo partial melting to produce basaltic magmas.
2. Mixtures of sedimentary rock and basaltic oceanic crust, such as those found in subduction zones, melt to form andesitic magmas.
3. Mixtures of sedimentary, igneous, and metamorphic continental crustal rocks melt to produce granitic magmas.

Thus, the mechanisms of magmatic differentiation must be much more complex than first recognized in a number of ways:

- Magmatic differentiation can begin with the partial melting of mantle and crustal rocks with a range of water contents over a range of temperatures.
- Magmas do not cool uniformly; they may exist transiently at a range of temperatures within a magma chamber. Differences in temperature within and among magma chambers may cause the chemical composition of magma to vary from one region to another.
- A few magma types are *immiscible*—they do not mix with one another, just as oil and water do not mix. When such magmas coexist in one magma chamber, each forms its own fractional crystallization series. Magmas that are *miscible*—that *do* mix—may follow a crystallization path different from that followed by any one magma type alone.

We now know more about the physical processes that interact with fractional crystallization within magma chambers (Figure 4.8). Magma at various temperatures in different parts of a magma chamber may flow turbulently, crystallizing as it circulates. Crystals may settle, then be caught up in currents again, and eventually be deposited on the chamber's walls. The margins of such a magma chamber may be a “mushy” zone of crystals and melt lying between the solid rock border of the chamber and the completely liquid magma within the heart of the chamber. And, at some mid-ocean ridges, such as the East Pacific Rise, a mushroom-shaped magma chamber may be surrounded by hot basaltic rock containing only small amounts (1 to 3 percent) of partial melt.

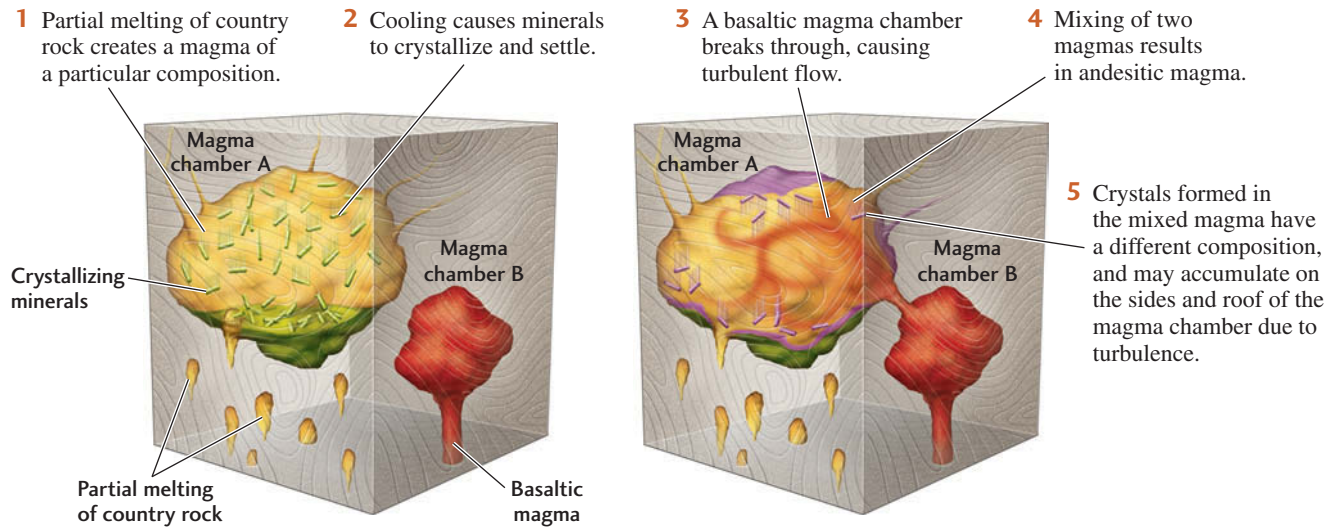


FIGURE 4.8 ■ Magmatic differentiation is a more complex process than first recognized. Some magmas derived from rocks of varying compositions may mix, whereas others are immiscible. Crystals may be transported to various parts of a magma chamber by turbulent flow in the liquid magma.

■ Forms of Igneous Intrusions

As noted earlier, we cannot directly observe the shapes of igneous intrusions. We can deduce their shapes and distributions only by observing parts of them where they have been uplifted and exposed by erosion, millions of years after the magma that formed them intruded and cooled.

We do have indirect evidence of current magmatic activity. Seismic waves, for example, have shown us the general outlines of the magma chambers that underlie some active volcanoes. In some nonvolcanic but tectonically active

regions, such as an area near the Salton Sea in Southern California, measurements in deep drill holes reveal crustal temperatures much hotter than normal, which may be evidence of a magmatic intrusion nearby. But these methods cannot reveal the detailed shapes or sizes of intrusions.

Most of what we know about igneous intrusions is based on the work of field geologists who have examined and compared a wide variety of outcrops and have reconstructed their histories. In the following pages, we consider some of these forms. **Figure 4.9** illustrates a variety of intrusive and extrusive structures.

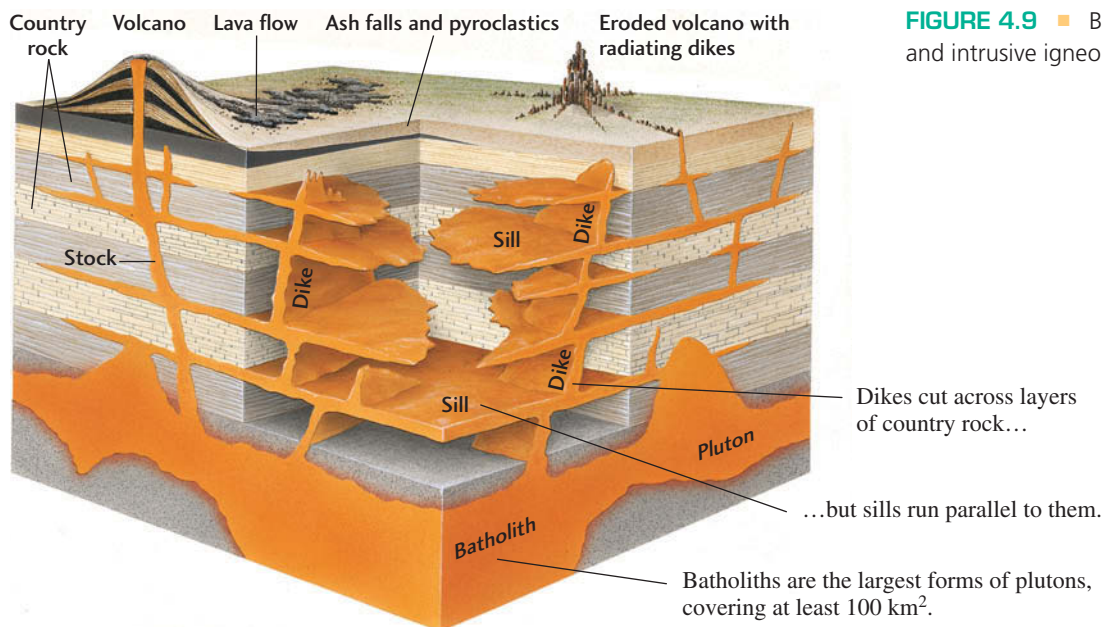


FIGURE 4.9 ■ Basic forms of extrusive and intrusive igneous structures.

Plutons

Plutons are large igneous bodies formed deep in Earth's crust. They range in size from a cubic kilometer to hundreds of cubic kilometers. We can study these large bodies when uplift and erosion uncover them or when mines or drill holes cut into them. Plutons are highly variable, not only in size but also in shape and in their relationship to the country rock.

This wide variation is due in part to the different ways in which magma makes space for itself as it rises through the crust. Most plutons intrude at depths greater than 8 to 10 km. At these depths, there are few holes or openings in the country rock because the pressure of the overlying rock would close them. But the upwelling magma overcomes even that great pressure.

Magma rising through the crust makes space for itself in three ways (Figure 4.10) that may be referred to collectively as *magmatic stoping*:

1. *Wedging open the overlying rock.* As the rising magma lifts the great weight of the overlying rock, it fractures that rock, penetrates the cracks, wedges them open, and so flows into the rock. Overlying rocks may bow upward during this process.
2. *Melting surrounding rock.* Magma also makes its way by melting country rock.
3. *Breaking off large blocks of rock.* Magma can push its way upward by breaking off blocks of the invaded crust. These blocks, known as *xenoliths* (from the Greek for “foreign rocks”), sink into the magma, where they may melt and blend into the liquid,

in some places changing the composition of the magma.

Most plutons show sharp zones of contact with country rock and other evidence of the intrusion of liquid magma into solid rock. Some plutons grade into country rock and contain structures vaguely resembling those of sedimentary rocks. The features of these plutons suggest that they formed by partial or complete melting of preexisting sedimentary rock.

Batholiths, the largest plutons, are great irregular masses of coarse-grained igneous rock that, by definition, cover at least 100 km² (see Figure 4.10). They are thick, horizontal, sheetlike or lobe-shaped bodies extending from a funnel-shaped central region. Their bottoms may extend 10 to 15 km deep, and a few are estimated to go even deeper. The coarse grain of batholiths results from slow cooling at great depths. Other, smaller plutons are called **stocks**. Both batholiths and stocks are **discordant intrusions**; that is, they cut across the layers of the country rock that they intrude.

Sills and Dikes

Sills and dikes are similar to plutons in many ways, but they are smaller and have a different relationship to the layering of the country rock (Figure 4.11). A **sill** is a sheetlike body formed by the injection of magma between parallel layers of bedded country rock. Sills are **concordant intrusions**; that is, their boundaries lie parallel to the country rock layers, whether or not those layers are horizontal. Sills range in thickness from a single centimeter to hundreds of meters, and they can extend over considerable areas. Figure 4.11a shows

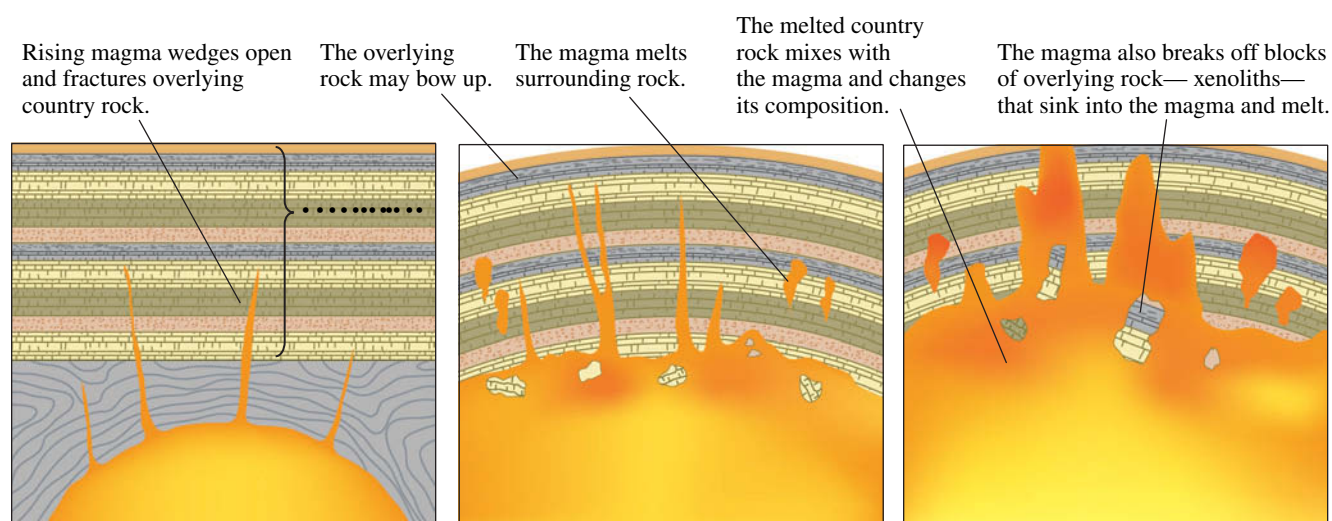


FIGURE 4.10 ■ Magmas make their way into country rock in three basic ways: by invading cracks and wedging open overlying rock, by melting country rock, and by breaking off pieces of rock. Pieces of broken-off country rock, called xenoliths, can become completely dissolved in the magma. If the country rock differs in composition from the magma, the composition of the magma may change.

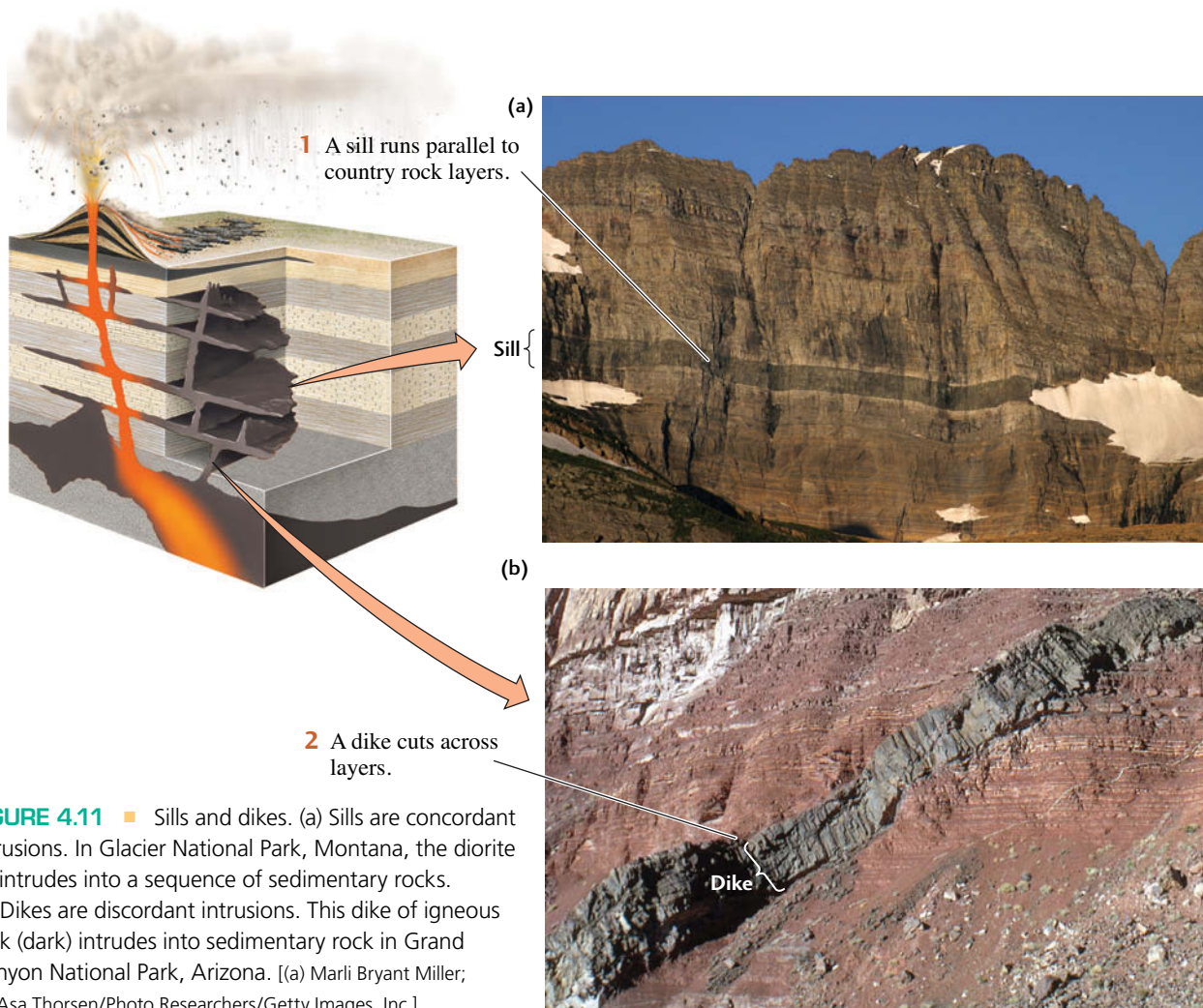


FIGURE 4.11 ■ Sills and dikes. (a) Sills are concordant intrusions. In Glacier National Park, Montana, the diorite sill intrudes into a sequence of sedimentary rocks. (b) Dikes are discordant intrusions. This dike of igneous rock (dark) intrudes into sedimentary rock in Grand Canyon National Park, Arizona. [(a) Marli Bryant Miller; (b) Asa Thorsen/Photo Researchers/Getty Images, Inc.]

a large sill at Finger Mountain, Antarctica. The 300-m-thick Palisades intrusion (see Figure 4.7) is another large sill.

Sills may superficially resemble lava flows and pyroclastic deposits, but they differ from those layers in four ways:

1. They lack the ropy, blocky, and vesicle-filled structures that characterize many volcanic rocks (see Chapter 12).
2. They are more coarse-grained than volcanic rocks because they have cooled more slowly.
3. Rocks above and below sills show the effects of heating: their color may have been changed or their mineral composition altered by contact metamorphism.
4. Many lava flows overlie weathered older flows or soils formed between successive flows; sills do not.

Dikes are the major route of magma transport in the crust. Dikes, like sills, are sheetlike igneous bodies, but dikes cut across the layers in bedded country rock (Figure 4.11b)

and so are discordant intrusions. Dikes sometimes form by forcing open existing fractures in the country rock, but more often they create channels through new cracks opened by the pressure of rising magma. Some individual dikes can be followed in the field for tens of kilometers. Their thicknesses vary from many meters to a few centimeters. In some dikes, xenoliths provide evidence of disruption of the country rock during the intrusion process. Dikes rarely exist alone; more typically, swarms of hundreds or thousands of dikes are found in a region that has been deformed by a large igneous intrusion.

The textures of dikes and sills vary. Many are coarse-grained, with an appearance typical of intrusive rocks. Many others are fine-grained and look much more like volcanic rocks. Because we know that texture corresponds to the rate of cooling, we can conclude that the fine-grained dikes and sills invaded country rock nearer Earth's surface, where the country rock was cold compared with the intrusions. Their fine texture is the result of rapid cooling. The coarse-grained ones formed at depths of many kilometers and invaded warmer rocks whose temperatures were much closer to their own.



FIGURE 4.12 ■ A granitic pegmatite vein. The center of the intrusion (*upper right*) cooled more slowly and developed coarser crystals. The margin of the intrusion (*lower left*) has finer crystals due to more rapid cooling. [John Grotzinger/Ramón Rivera-Moret/Harvard Mineralogical Museum.]

Veins

Veins are deposits of minerals found within a rock fracture that are foreign to the country rock. Irregular pencil-shaped or sheet-shaped veins branch off from the tops and sides of many igneous intrusions. They may be a few millimeters to several meters across, and they tend to be tens of meters to kilometers long or wide. The formation of veins is described in more detail in Chapter 3.

Veins of extremely coarse grained granite cutting across much finer grained country rock are called **pegmatites** (Figure 4.12). Pegmatites crystallize from a water-rich magma in the late stages of solidification.

Other veins are filled with hydrous minerals that are known to crystallize from hydrothermal solutions. From laboratory experiments, we know that these minerals typically crystallize at temperatures of 250°C to 350°C—high temperatures, but not nearly as high as the temperatures of magmas. The solubility and composition of the minerals in these hydrothermal veins indicate that abundant water was present as the veins formed. Some of the water may have come from the magma itself, but some may have been underground water in the cracks and pore spaces of the intruded rocks. On land, groundwaters originate as rainwater seeps into the soil and surface rocks. Hydrothermal veins are also abundant along mid-ocean ridges, where seawater infiltrates cracks in the newly formed seafloor, circulates down into hotter regions of the ridge, and reemerges at hydrothermal vents in the rift valley between the spreading plates. Hydrothermal processes at mid-ocean ridges are examined in more detail in Chapter 12.

■ Igneous Processes and Plate Tectonics

Geologists have observed that the facts and theories of igneous rock formation fit nicely into a framework based on plate tectonic theory. The geometry of plate movements is the link we need to tie tectonic activity and rock composition to igneous processes (Figure 4.13). Batholiths, for example, are found in the cores of many mountain ranges formed by the convergence of two plates. This observation implies a connection between pluton formation and the mountain-building process, and between both of those processes and plate movements. Similarly, our knowledge of the temperatures and pressures at which different kinds of rock melt gives us some idea of where melting takes place. For example, we know that mixtures of sedimentary rocks, because of their composition and water content, should melt at temperatures several hundred degrees below the melting point of basalt. This information leads us to predict that basalt will start to melt near the base of the crust in tectonically active regions of the upper mantle and that sedimentary rocks will melt at shallower depths.

Magma forms most abundantly in two plate tectonic settings: mid-ocean ridges, where two plates diverge and the seafloor spreads, and subduction zones, where one plate dives beneath another. Mantle plumes, though not associated with plate boundaries, also produce large amounts of magma.

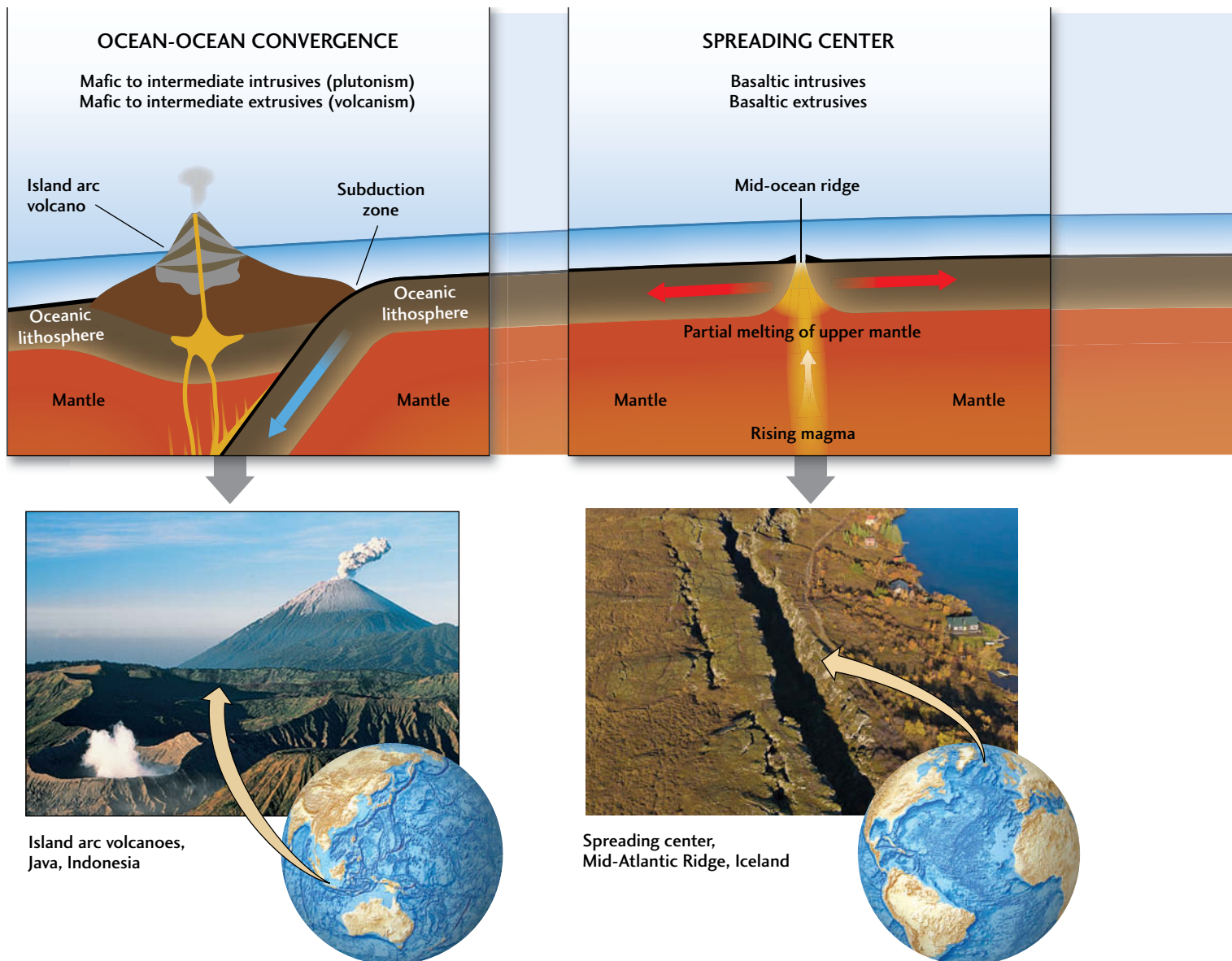


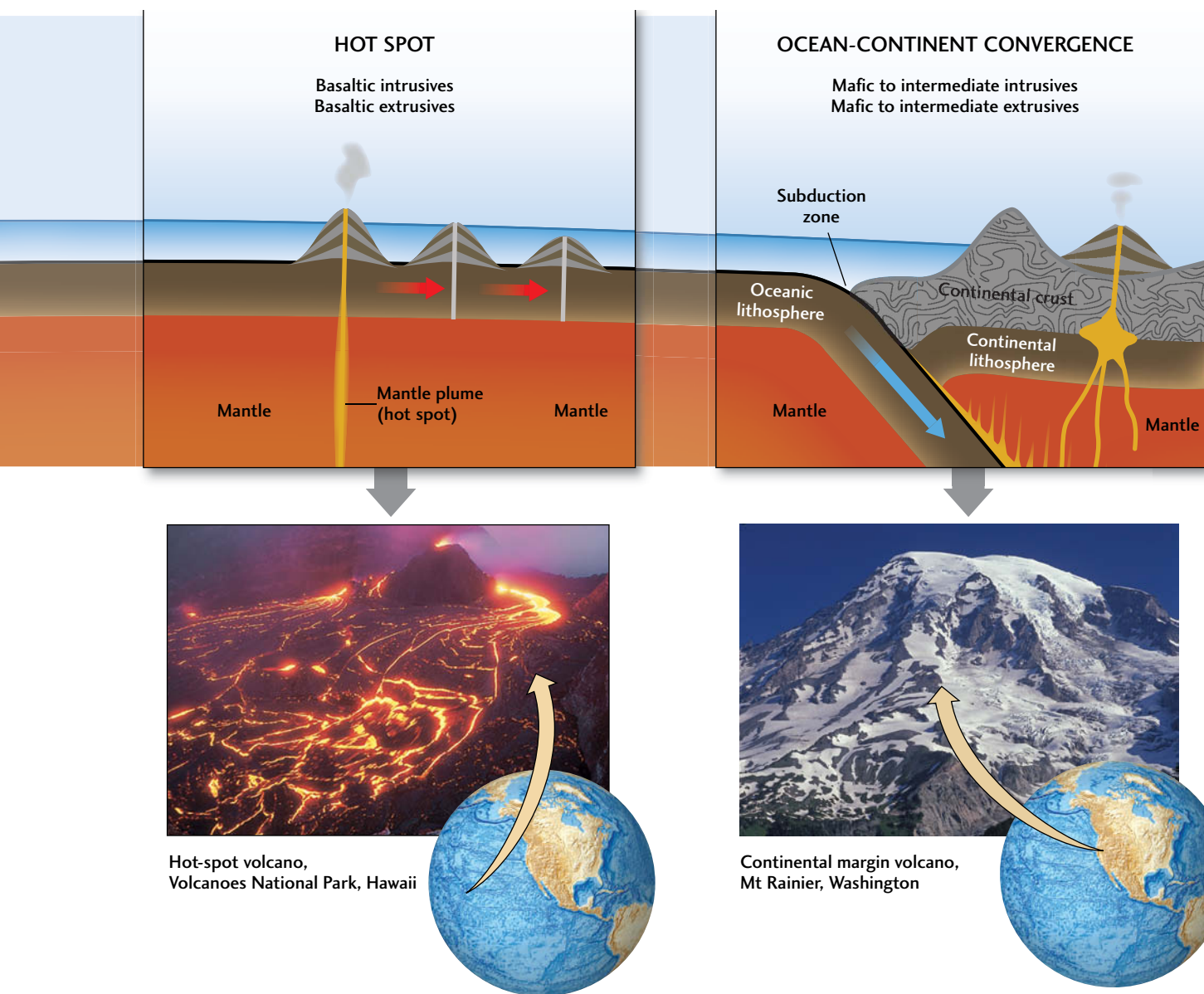
FIGURE 4.13 ■ Magmatic activity is related to plate tectonic settings. [Photos (left to right): Mark Lewis/Stone/Getty Images; Ragnar Th Sigurdsson/© ARCTIC Images/Alamy; G. Brad Lewis/Stone/Getty Images; © Michael Sedam/Age Fotostock.]

Spreading Centers as Magma Factories

Most igneous rocks are formed at mid-ocean ridges by the process of seafloor spreading. Each year, approximately 19 km^3 of basaltic magma is produced along the mid-ocean ridges in the process of seafloor spreading—a truly enormous volume. In comparison, all the active volcanoes along convergent plate boundaries (about 400) generate volcanic rock at a rate of less than $1 \text{ km}^3/\text{year}$. Enough magma has erupted during seafloor spreading over the past 200 million years to create all of the present-day seafloor, which covers nearly two-thirds of Earth's surface. Throughout the mid-ocean ridge network, decompression melting of mantle material that wells up along rising convection currents

creates magma chambers below the ridge axis. These magmas are extruded as lavas and form new seafloor. At the same time, intrusions of gabbro are emplaced at depth.

Before the advent of plate tectonic theory, geologists were puzzled by unusual assemblages of rocks that were characteristic of the seafloor but were found on land. Known as **ophiolite suites**, these assemblages consist of deep-sea sediments, submarine basaltic lavas, and mafic igneous intrusions (Figure 4.14). Using data gathered from deep-diving submarines, dredging, deep-sea drilling, and seismic exploration, geologists now explain these rocks as fragments of oceanic lithosphere that were transported by seafloor spreading and then raised above sea level and thrust onto a continent in a later episode of plate collision. On some of the more complete ophiolite suites preserved on



land, we can literally walk across rocks that used to lie along the boundary between Earth's oceanic crust and mantle.

How does seafloor spreading work? We can think of a spreading center as a huge factory that processes mantle material to produce oceanic crust. **Figure 4.15** is a highly schematic and simplified representation of what may be happening, based in part on studies of ophiolite suites found on land and on information gleaned from deep-sea drilling and seismic profiling. Deep-sea drilling has penetrated to the gabbro layer of the seafloor, but not to the crust-mantle boundary below. Seismic profiling has found several small magma chambers similar to the one shown in **Figure 4.15**.

INPUT MATERIAL: PERIDOTITE IN THE MANTLE The raw material fed into this magma factory comes from the convecting asthenosphere, in which the dominant rock type is peridotite. The mineral composition of the average

peridotite in the mantle is chiefly olivine, with smaller amounts of pyroxene and garnet. Temperatures in the asthenosphere are hot enough to melt a small fraction of this peridotite (less than 1 percent), but not hot enough to generate substantial volumes of magma.

PROCESS: DECOMPRESSION MELTING Decompression melting is the process that generates great volumes of magma from peridotite at spreading centers. Recall that a decrease in pressure generally lowers a mineral's melting temperature. As the plates pull apart, the partially molten peridotite is sucked inward and upward toward the spreading center. The decrease in pressure as the peridotite rises causes a large fraction of the rock (up to 15 percent) to melt. The buoyancy of the melt causes it to rise faster than the denser surrounding rock. This process separates the liquid rock from the remaining crystal mush to produce large volumes of magma.

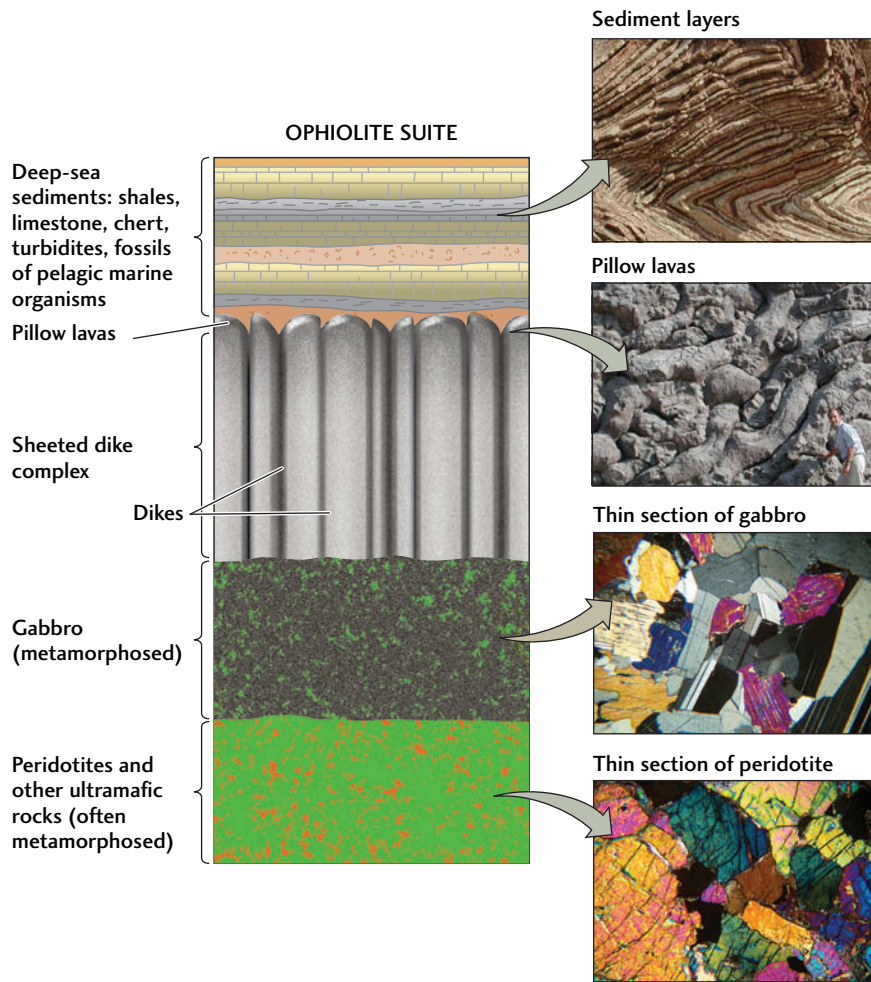


FIGURE 4.14 ■ Idealized section of an ophiolite suite. The combination of deep-sea sediments, pillow lavas, sheeted dikes of gabbro, and mafic igneous intrusions indicates a deep-sea origin. [Photos courtesy of John Grotzinger. Thin sections courtesy of T. L. Grove.]

OUTPUT MATERIAL: OCEANIC CRUST PLUS MANTLE LITHOSPHERE The peridotites subjected to this process do not melt evenly: the garnet and pyroxenes they contain melt at lower temperatures than the olivine. For this reason, the magma generated by decompression melting is not peridotitic in composition; rather, it is enriched in silica and iron. This basaltic melt forms a magma chamber below the mid-ocean ridge crest, where it separates into three layers (see Figure 4.15):

1. Some magma rises through the narrow cracks that open where the plates are separating and erupts into the ocean, forming the basaltic *pillow lavas* that cover the seafloor.
2. Some magma solidifies in the cracks as vertical, sheeted dikes of gabbro.
3. The remaining magma solidifies as massive intrusions of gabbro as the underlying magma chamber is pulled apart by seafloor spreading.

These igneous units—pillow lavas, sheeted dikes, and massive gabbros—are the basic layers of the crust that geologists have found throughout the world's oceans.

Seafloor spreading results in another layer beneath this oceanic crust: the residual peridotite from which the basaltic magma was originally derived. Geologists consider this layer to be part of the mantle, but its composition is different from that of the convecting asthenosphere. In particular, the extraction of the basaltic melt makes the residual peridotite richer in olivine and stronger than ordinary mantle material. Geologists now believe it is this olivine-rich layer at the top of the mantle that gives the oceanic lithosphere its great rigidity.

Above the pillow lavas, a blanket of deep-sea sediment begins to cover the newly formed oceanic crust. As the seafloor spreads, these layers of sediment, pillow lavas, dikes, and gabbro are transported away from the mid-ocean ridge where this characteristic sequence of rocks is assembled, almost as if they were moving along a production line.

Subduction Zones as Magma Factories

Other types of magmas underlie regions where volcanoes are highly concentrated, such as the Andes of South America and the Aleutian Islands of Alaska. Both of these

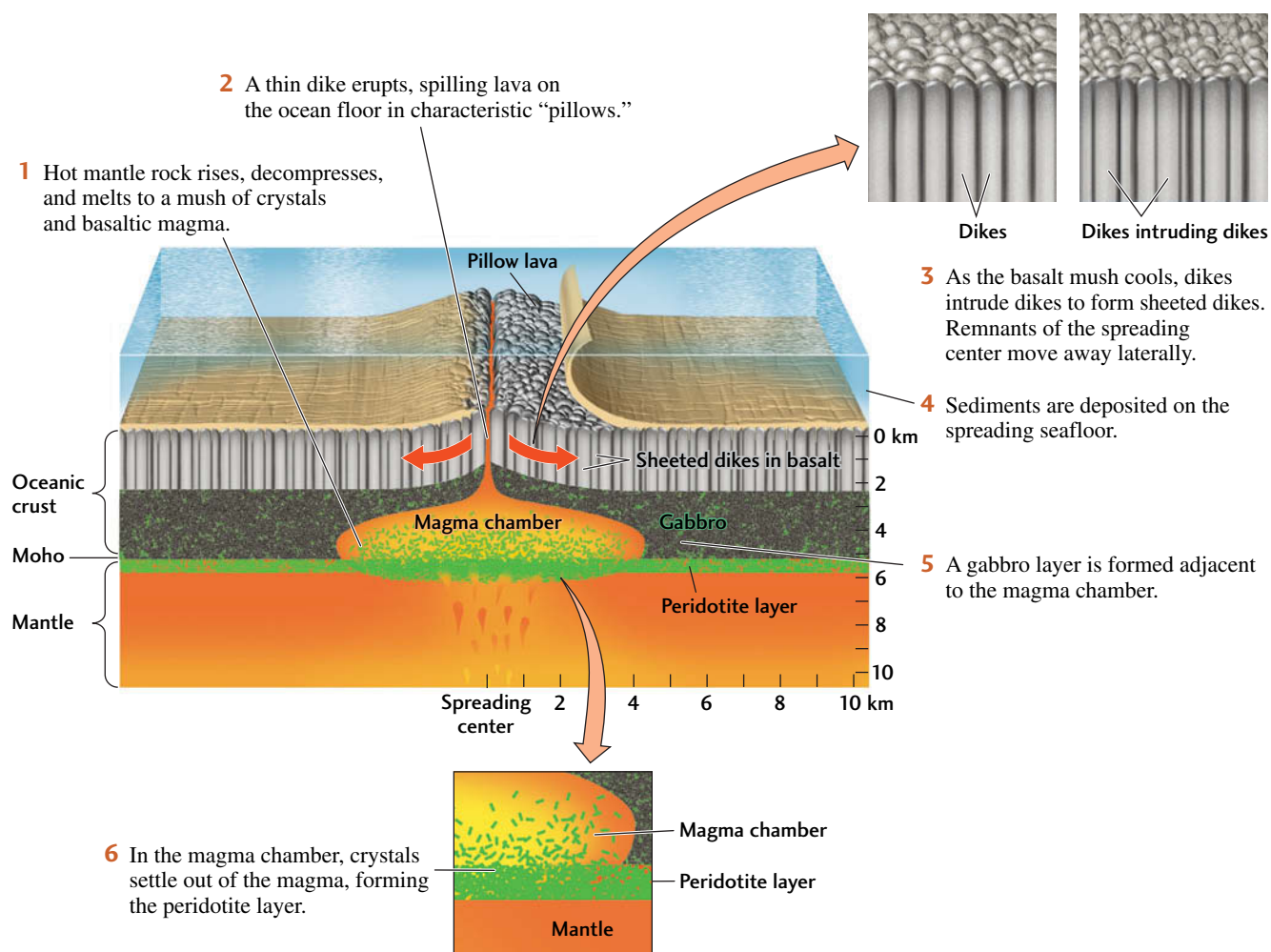


FIGURE 4.15 ■ Decompression melting creates magma at seafloor spreading centers.

regions lie over subduction zones, which are major magma factories (**Figure 4.16**). They generate magmas of varying composition, depending on how much and what kinds of materials are subducted.

Where oceanic lithosphere is subducted beneath a continent, the resulting volcanoes and volcanic rocks form a volcanic mountain belt on the continent. The Andes, which mark the subduction of the Nazca Plate beneath South America, are one such mountain belt. Similarly, subduction of the small Juan de Fuca Plate beneath western North America has generated the Cascade Range, with its active volcanoes, in northern California, Oregon, and Washington. Where oceanic lithosphere is subducted beneath oceanic lithosphere, a deep-sea trench and a volcanic island arc are formed.

INPUT MATERIAL: A MIXED BAG The variation in the chemical and mineral composition of magmas at subduction zones is a clue that the magma factories at convergent boundaries operate differently from those at spreading centers. The raw materials for these magma factories include mixtures of seafloor sediments, mixtures of basaltic

oceanic crust and felsic continental crust, mantle peridotite, and water.

PROCESS: FLUID-INDUCED MELTING The basic mechanism of magma formation at subduction zones is fluid-induced melting. The fluid involved is primarily water, which, as we have seen, lowers the melting temperature of rock. By the time oceanic lithosphere is subducted at a convergent plate boundary, a lot of water has been incorporated into its outer layers. We have already mentioned one of the processes responsible: hydrothermal activity during the formation of oceanic lithosphere. Some of the seawater that circulates through the crust near a spreading center reacts with basalt to form new hydrous minerals. In addition, as the lithosphere ages and is transported across the ocean basin, sediments containing water are deposited on its surface. The rocks formed from these sediments include shales, which contain large proportions of hydrous clay minerals. Some of these sediments get scraped off the subducting plate at the deep-sea trench, but much of this water-laden material is carried downward into the subduction zone.

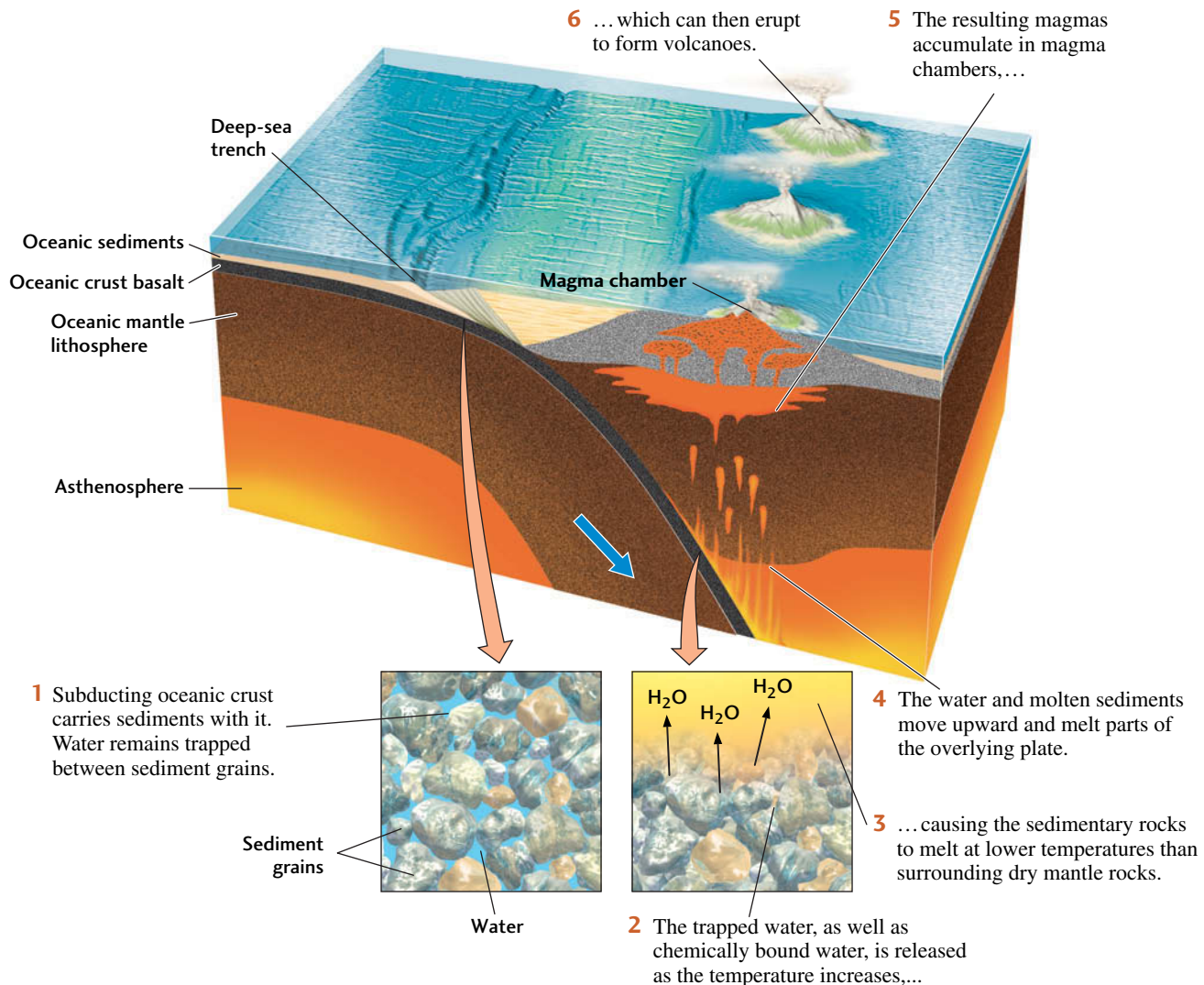


FIGURE 4.16 ■ Fluid-induced melting creates magma in subduction zones.

As the lithospheric slab moves downward, it is subjected to increasing pressure. Water is squeezed out of the minerals in the outer layers of the descending crust and rises buoyantly into the mantle above the descending crust. At moderate depths of about 5 km, the temperature increases to about 150°C. Here, more water is released by metamorphic chemical reactions as basalt is converted to *amphibolite*, which is composed of amphibole and plagioclase feldspar (see Chapter 6). As other chemical reactions take place, additional water is released at depths ranging from 10 to 20 km. Finally, at depths greater than 100 km, the temperature increases to 1200°C–1500°C, and the subducted slab undergoes an additional metamorphic transition induced by the increased pressure. Amphibolite is converted to *eclogite*, which is composed of pyroxene and garnet (see Chapter 6). Here, the increase in both pressure and temperature in

the subducting slab releases all of its remaining water in addition to other materials.

During subduction, the released water induces melting in the descending basalt-rich oceanic crust and in the overlying peridotite-rich mantle material. Most of the resulting mafic magma accumulates at the base of the crust of the overriding plate, and some of it intrudes into the crust to form magma chambers, resulting in the formation of volcanoes.

OUTPUT: MAGMAS OF VARYING COMPOSITION

The magmas produced by fluid-induced melting at subduction zones are essentially basaltic, although their composition is more variable than that of mid-ocean ridge basalts. The composition of these magmas is further altered during their residence in the crust. Within magma chambers, the process of fractional crystallization increases the magma's silica content, producing eruptions of andesitic

lavas. Where the overlying plate is continental, the heat from the magmas can melt the felsic rocks in the crust, forming magmas with an even higher silica content, such as dacitic and rhyolitic magmas. The contribution of lithospheric fluids to these magmas is suggested by the presence in the magmas of trace elements known to be present in oceanic crust and sediments.

Mantle Plumes as Magma Factories

Mantle plumes, like spreading centers, are sites of decompression melting, but they differ from spreading centers by forming within lithospheric plates rather than along the margins of plates. These plumes of hot mantle material rise from deep within Earth, possibly as deep as the core-mantle boundary. Mantle plumes that reach the surface, most of them far from plate boundaries, form the “hot spots” of Earth. At these locations, basaltic magmas produced by decompression melting of mantle material may erupt in huge outpourings to form islands, such as the Hawaiian Islands, or basalt plateaus, such as the Columbia Plateau in the Pacific Northwest of North America. Mantle plumes and hot spots are discussed in more detail in Chapter 12.

SUMMARY

How are igneous rocks classified? Igneous rocks can be divided into two broad textural classes: coarse-grained rocks, which are intrusive and therefore cooled slowly; and fine-grained rocks, which are extrusive and cooled rapidly. Igneous rocks can also be classified on the basis of their silica content using a scale that runs from felsic (rich in silica) to ultramafic (poor in silica).

KEY TERMS AND CONCEPTS

andesite (p. 96)
 basalt (p. 96)
 batholith (p. 103)
 bomb (p. 93)
 concordant intrusion (p. 103)
 country rock (p. 93)
 dacite (p. 96)
 decompression melting (p. 98)
 dike (p. 104)
 diorite (p. 96)

discordant intrusion (p. 103)
 extrusive igneous rock (p. 93)
 felsic rock (p. 95)
 fluid-induced melting (p. 98)
 fractional crystallization (p. 99)
 gabbro (p. 96)
 granite (p. 95)
 granodiorite (p. 96)
 intermediate igneous rock (p. 96)

How and where do magmas form? Magmas form at places in the lower crust and mantle where temperatures are high enough for partial melting of rock. Because the minerals within a rock melt at different temperatures, the composition of magmas varies with temperature. Pressure raises the melting temperature of rock, and the presence of water lowers it. Because melted rock is less dense than solid rock, magma rises through the surrounding rock, and drops of magma come together to form magma chambers.

How does magmatic differentiation account for the variety of igneous rocks? Because different minerals crystallize at different temperatures, the composition of magma changes as it cools and various minerals are withdrawn by crystallization.

What are the forms of igneous intrusions? Large intrusive igneous bodies are called plutons. The largest plutons are batholiths, which are thick horizontal masses extending from a funnel-shaped central region. Stocks are smaller plutons. Less massive than plutons are sills, which lie parallel to the layers of bedded country rock, and dikes, which cut across those layers. Veins form where water is abundant, either in the magma or in the surrounding country rock.

How do plate tectonic processes affect magma production? Magmas are produced at two types of plate boundaries. At spreading centers, peridotite rises from the mantle and undergoes decompression melting to form basaltic magma. At subduction zones, subducting oceanic lithosphere undergoes fluid-induced melting to generate magmas of varying composition. Mantle plumes within lithospheric plates are also sites of decompression melting that produce basaltic magmas.

intrusive igneous rock (p. 93)
 lava (p. 92)
 mafic rock (p. 96)
 magma chamber (p. 99)
 magmatic differentiation (p. 99)
 obsidian (p. 94)
 ophiolite suite (p. 106)
 partial melting (p. 98)
 pegmatite (p. 105)
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